2017 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.312 Languages and Compilation

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a You've been employed as an external consultant by *Amazon* to design a new customer database for their new *beer home delivery* service, called *Quench*. One of the key tasks is to import name and address information from their existing business, so you need to define a formal language in order to check the integrity of the existing information. Write a set of grammar rules in Backus-Naur Form (BNF) to define each *address* element of the current customer database. You should write BNF rules describing the structure of one address entry and how it breaks down into house name or number, street name, town, and city. You should include at least one optional item such as county and/or postcode in your BNF rules. You need not specify a full set of terminal symbols corresponding to each non-terminal; two or three example terminal symbols are sufficient for each non-terminal. You do not need to include punctuation characters in your grammar.

[7 marks]

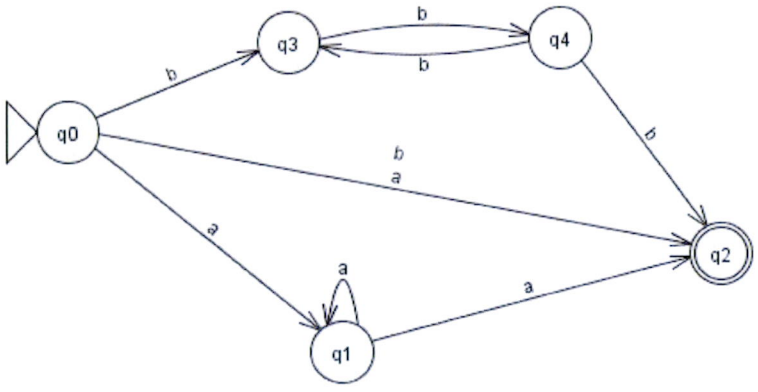
1.b Using your BNF grammar defined in part (a), draw a parse tree for following address:

999 Letsby Avenue
Pincham
Surrey
SR2 5XX

[4 marks]

1.c Consider the following finite state recognizer (FSR). Write down the list of non-deterministic states in this machine.

[2 marks]



Question 1 continues on next page...

Question 1 continued...

1.d Using the *subset construction algorithm*, convert the non-deterministic FSR in part (c) to a deterministic FSR. You should describe the conversion step by step showing which states you are grouping together to form new states. You may relabel the final states if you wish, but make it clear what states in the original correspond to the new states. Six marks will be awarded for describing the conversion process and four marks will be awarded for the final diagram.

[10 marks]

1.e Now, taking your deterministic FSR from part (d), express the language accepted as a regular expression.

[2 marks]

Total 25 marks

Question 2

2.a Name three of the possible types of grammar in the Chomsky hierarchy and for each one describe the general form of the production rules and list any restrictions on them.

[6 marks]

2.b Consider the following grammar:

$$S \rightarrow bA \mid aB$$
$$A \rightarrow a \mid aS \mid bAA$$
$$B \rightarrow b \mid bS \mid aBB$$

Classify this grammar according to the Chomsky hierarchy. Illustrate by example and with reference to your answer for part (a) why this grammar is of a certain type and also why it is not of the other types.

[3 marks]

2.c Show that the grammar in part (b) is ambiguous. Illustrate your answer by providing two possible parse trees for the sentence: bbaaba

[4 marks]

2.d Describe in words the format of sentences that are generated or recognised by the grammar in part (b) and give two example sentences.

[2 marks]

2.e Generate a *non-deterministic* push-down automata *directly* from the grammar in part (b).

[6 marks]

2.f Define the *halting problem* and briefly explain (maximum one paragraph) the implications of the halting problem for a Universal Turing Machine when it is simulating a Turing Machine.

[4 marks]

Total 25 marks

Question 3

3.a Explain the advantages of having an Intermediate Language (IL) stage in a compiler suite.

[3 marks]

3.b Explain how an Intermediate Language (IL) stage potentially limits the functionality of a compiler?

[4 Marks]

For questions **3.c** and **3.d** the following grammar and input stream are used.

E -> V
E -> V O E
O -> "+" | "-" | "*" | "/"
V -> An Unsigned Integer

Input => [1], [+], [2], [*], [4]

3.c Show at each step the state of a stack-based expression interpreter using the grammar defined above, when given the input token stream also defined above.

[15 Marks]

3.d What answer does this expression interpreter produce from the input string above, and what might be incorrect about this result?

[3 Marks]

Total 25 marks

Question 4

4.a Define, and given an example of explicit and implicit type conversion.

[4 Marks]

4.b In a language using *structure equivalence* to determine type equality, explain how might the following pseudocode definition cause problems.

```
type node is record
   value : integer
   next : pointer to node
end record
```

[4 Marks]

4.c Here is a grammar G, with terminals {v, m, t}, non-terminals (Z, Q, R} and Z is the distinguished symbol.

Z -> Q v R
Z -> R
Q -> m R
R -> t

Build the Action and Goto tables for this grammar using the procedure below, as taught in the lectures:

Step 1: Augment the grammar. [1 Mark]
Step 2: Build the item sets. [8 Marks]
Step 3: Generate the state transition table. [6 Marks]
Step 4: Produce the Action and Goto Tables. [2 Marks]

[17 Marks]

Total 25 marks

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